JISHNU AURO GHOSH

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SUMMARY

Industrial Engineering senior at Purdue with hands-on experience in production scheduling, systems optimization, and data-driven operations. Built Python-based scheduling models, enterprise dashboards, and analytics pipelines across manufacturing and research environments. Effective collaborator and technical lead with experience managing multi-person engineering projects.

TECHNICAL SKILLS

IE / Operations: production scheduling, capacity analysis, bottleneck analysis, work measurement, labor planning, facility layout/routing, standard work

Optimization / Analytics: linear programming, operations research, stochastic modeling, regression, hypothesis testing, data visualization

Programming: Python, SQL, R, MATLAB, JavaScript, PHP, Excel, Jupyter, Git/GitHub, VS Code, Tableau

Libraries / Solvers: pandas, NumPy, matplotlib, scikit-learn, TensorFlow, Gurobi

EDUCATION

Purdue University — B.S. Industrial Engineering | Minor: Law & Society *Aug 2022 – May 2026*

Coursework: Operations Research; Linear Programming; Integrated Production Systems I/II; Work Analysis & Design I/II; Manufacturing Processes; Industrial Control Systems; Engineering Economics

Leadership: *Founder, Chi Psi Fraternity (Alpha Epsilon Tau); Service Chair, 2022–2023*

ENGINEERING, ANALYTICS & OPERATIONS EXPERIENCE

Production Scheduling Capstone · Major Aerospace OEM · Purdue IE Capstone *Spring 2026*

- Led code development and Git/GitHub workflow for a five-person team building a production scheduler for post-final-assembly movement through Final Goods Inventory and delivery operations.
- Modeled asset state, location occupancy, paint/inspection workflows, delivery readiness, labor-driven task completion, and layout-based move feasibility in Python.
- Integrated staged Excel inputs, move-time matrices, trace exports, labor allocation outputs, and KPI scheduler tables for scenario planning through May 2027; placed 3rd among IE capstone teams.

Enterprise Systems Dashboard · Purdue · Operations & Systems Project *Fall 2025*

- Built SQL/PHP/JavaScript enterprise dashboard with role-based access controls and KPI views for senior-manager decision support across multi-department operations.
- Designed normalized data schemas and optimized query logic to surface operational metrics and enable real-time visibility into cross-functional workflows.

Research Assistant — Human Integration Lab · Purdue School of Industrial Engineering *2023 – 2024*

- Supported human-systems research on teamwork and situational awareness using structured workflows, fNIRS, eye-tracking, and biometric sensors in VR construction scenarios.
- Built Python/Excel scoring pipelines converting raw experimental data into performance and situational-awareness measures across 160+ hours of sessions.

Margo Katherine Wilke Undergraduate Researcher · Purdue College of Liberal Arts *2023 – 2025*

- Built longitudinal text dataset and reproducible Python analysis workflows using hypothesis testing and regression; translated interdisciplinary research questions into statistical pipelines.

SELECTED TECHNICAL PROJECTS

Optimization Algorithm for Hydrological Modeling · Purdue Technical Project *Fall 2025*

- Implemented and benchmarked LP solvers — Simplex and Interior Point methods — with degeneracy handling and anti-cycling safeguards; compared convergence behavior across problem sets.